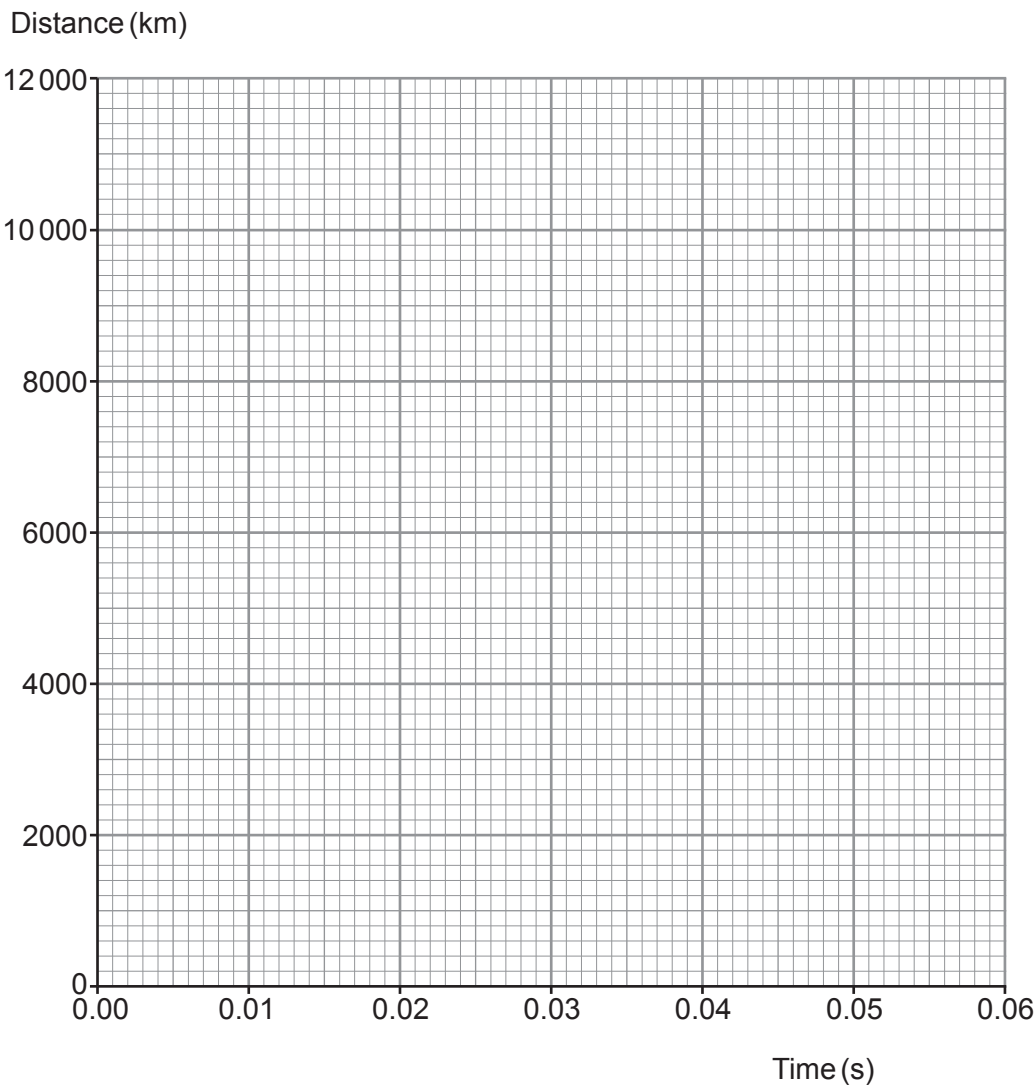


7. Communication between the UK and USA can be made by optical fibres or by geostationary satellites.

The table below shows the time taken by infra-red waves to travel different distances through an optical fibre.

Distance (km)	2000	4000	6000	8000	10 000
Time (s)	0.01	0.02	0.03	0.04	0.05

(a) Use the data to plot a graph on the grid below and draw a suitable straight line. [3]



- (b) An infra-red signal sent from the UK to the USA by optical fibre travels 9000 km.
A microwave signal sent from the UK to the USA by geostationary satellite travels 72 000 km and takes 0.24 s.
Chris thinks it is quicker to send signals by geostationary satellite.

Use a value from the graph to explain whether you agree with Chris.

[2]

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5

